Min-En Chou (Elsa)

Experimental AMO physics • Optical engineering • Scientific visualization

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RESEARCH INTERESTS

Autonomous ultracold-atom systems · AI-driven research and optical design · Neutral-atom platforms
Quantum resource estimation · Quantum error correction · Seeing quantum phenomena

EDUCATION

2018 – 2020 **M.S. in Physics**, National Tsing Hua University, Hsinchu, Taiwan
Thesis: *"Building a compact apparatus for the creation of ultracold Li and Cs atoms."*
Advisor: Shih-Kuang Tung

2014 – 2018 **B.S. in Physics**, National Tsing Hua University, Hsinchu, Taiwan
Graduated in the top 10% of the class. Academic Excellence Award, 2014 and 2015.

RESEARCH EXPERIENCE

2025 – 2026 **Research Assistant**
Future Narratives Lab, National Yang Ming Chiao Tung University, Hsinchu, Taiwan
- Built quantum HCI for a VR qubit art system; co-authored a SIGGRAPH poster paper [4].
- Organized workshops and a cross-lab symposium; managed lab operations and reporting.

2017 – 2020 **Undergraduate & Graduate Researcher**
Quantum Gas Laboratory, National Tsing Hua University, Hsinchu, Taiwan · Advisor: Shih-Kuang Tung
- Co-built a dual-species ^{7}Li–^{133}Cs machine from an empty optical table to a working MOT.
- Developed a compact 2D-MOT source that replaces a Zeeman slower for both species [3].

2016 – 2017 **Undergraduate Researcher**
National Tsing Hua University, Hsinchu, Taiwan · Advisor: Chung-Yu Mou
Numerical study of magnetic monopole phenomena in topological insulators.

INDUSTRY EXPERIENCE

2023 – 2025 **Optical R&D Engineer**
LARGAN Precision Co., Ltd., Taichung, Taiwan
- Designed and toleranced imaging lenses for tier-1 brands; inventor on three US filings.
- Developed image processing algorithms for ISP and image quality evaluation of lenses.
- Built a parallelized design workflow and modeling method for non-circular light shield.

2021 – 2022 **Patent Engineer**
TSAI, LEE & CHEN / JC Patent Attorneys Office, Taipei, Taiwan
Patent drafting and patentability analysis in display, semiconductor, robotics, and AR optics.

PUBLICATIONS

[4] 2026 Chang, H. C., Lang, H. C., Liu, I. C. H. & **Chou, M. E.** "Gestural control of quantum information in virtual reality." *In SIGGRAPH Posters '26. Article 48, pp. 1–3*.

[3] 2021 Chen, Y. D., Li, W. X., **Chou, M. E.**, Kuo, C. H., Li, C. S. & Tung, S. "Cold lithium–cesium slow atomic beam from a two-dimensional magneto-optical trap." *Physical Review A* **103**, 023102.

[2] 2020 Chen, Y. D., Li, W. X., Li, C. S., **Chou, M. E.** & Tung, S. K. "Towards a dual-species Bose–Einstein condensate of ^{7}Li and ^{133}Cs." *APS DAMOP Meeting*.

[1] 2018 Chen, Y. D., Li, W. X., Li, C. S., Chen, A. L., **Chou, M. E.**, Kuo, C. H., Jhang, C. Y. & Tung, S. K. "A new lithium–cesium machine for the study of quasiparticle localization in Bose–Einstein condensates." *APS DAMOP Meeting*.

PATENTS

2026 **Chou, M. E.** & Liao, G. J. "Imaging optical lens system, imaging apparatus and electronic device." US 2026/0072251 A1 (LARGAN Precision), pending.

TEACHING

2017 **Teaching Assistant**
General Physics Laboratory, National Tsing Hua University, Hsinchu, Taiwan
Guided undergraduates through experiments and graded lab reports.

2015 **Lecturer**
High School Science Camp, National Chiayi Senior High School · Chiayi, Taiwan
Taught a lecture introducing thermal and statistical physics to high school students.

SKILLS

Atomic physics	Laser cooling, 2D/3D MOTs, UHV assembly, fiber coupling, polarization optics, field-coil
Optics	Lens design and tolerancing (CODE V), stray-light analysis (ASAP), image processing
Computation	Python, MATLAB, C#, Inventor CAD, 3D printing
Interactive	Unity, JavaScript, HTML/CSS, LaTeX
Languages	Mandarin (native), English

Last updated September 2026